Rational square roots

Animath — Stage olympique de Sophia-Antipolis · 2006 · France

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Show that if a, b and $\sqrt{a}+\sqrt{b}$ are rational, then so are $\sqrt{a}$ and $\sqrt{b}$.
Can one deduce from the rationality of a, b, c and $\sqrt{a}+\sqrt{b}+\sqrt{c}$ that of $\sqrt{a}$, $\sqrt{b}$ and $\sqrt{c}$?